

ETSUBDINK ABREHAM

Software Engineering Student | Full-Stack & AI Enthusiast

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SUMMARY

Software Engineering student at Adama Science and Technology University (graduating June 2026) with hands-on experience in full-stack development, REST APIs, real-time systems, and database design. Strong foundation in data structures, algorithms, debugging, and testing. Seeking to apply technical skills to AI evaluation and software engineering research.

TECHNICAL SKILLS

Languages:	Python • JavaScript • C# • Java • SQL • HTML/CSS
Frameworks:	ASP.NET Core • React • TypeScript • REST APIs • SignalR • Selenium
Databases:	MySQL • SQL Server • Database Design • Data Modeling
CS Fundamentals:	Data Structures & Algorithms • OOP • Debugging • Testing • Software Engineering Principles
Tools & AI:	Git • GitHub • Postman • VS Code • MQTT • ML Fundamentals • Cloud Fundamentals

PROJECTS

Smart Health Kiosk System Final Year Project	Python • ASP.NET • MQTT • REST API • MySQL
- Architected a full-stack healthcare kiosk with remote patient monitoring, appointment scheduling, and AI-assisted health insights. - Engineered secure MQTT communication with TLS encryption and token-based authentication for kiosk-to-cloud data transmission. - Built RESTful APIs for subscription and payment modules; modeled a normalized SQL Server schema with multi-role access and query optimization.	
SaaS Hotel Management Platform	ASP.NET Core • React • TypeScript • SQL Server • Stripe API
- Built a multi-tenant SaaS platform for hotel reservations, room inventory, staff operations, and payments via a centralized dashboard. - Engineered REST APIs with ASP.NET Core and role-based access control; integrated Stripe for subscription billing and transaction management. - Designed responsive React/TypeScript dashboards and a scalable SQL Server schema with tenant data isolation and clean architecture.	
Ethiopian Bank Exchange Rate Aggregator	C# • ASP.NET • SignalR • HtmlAgilityPack • SQL Server
- Built an ASP.NET scraping platform using HtmlAgilityPack to aggregate live exchange rate data from 10+ Ethiopian commercial banks. - Integrated SignalR for real-time dashboard updates; designed a normalized SQL Server database for historical rate analysis and trend comparison.	

EDUCATION

B.Sc. Software Engineering — Adama Science and Technology University (ASTU)	Expected June 2026
Relevant Coursework: Data Structures & Algorithms • Database Systems • Software Testing & QA • Machine Learning • Operating Systems • Computer Networks	

RESEARCH INTERESTS

AI Privacy & Data Protection • ML Model Evaluation • Screenshot Detection Systems • Secure Messaging • Algorithmic Problem Solving • AI-Assisted Software Engineering